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# Oceans, hormones and female flow in *Twelfth Night*

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**ABSTRACT**
Although a shipwreck in *Twelfth Night* violently separates Viola from her twin, the catastrophe also shows Shakespeare's audience that water is a tool which can assuage grief and allow Viola to hide herself while operating as an instrument for managing other women's bodies and desires, too. Disguising herself as a eunuch allows Viola to dam up the flood of desire and halt growth, but these infrastructural adjustments to flow work on larger and smaller scales, reorganising the imagination while upending how the world is viewed. Power operates differently when water swamps thinking and *Twelfth Night*'s multiple references to things like 'tears', 'brine', 'foam', 'waves', and 'storms' help us see water as a symbol, resource and ordering device. If women's bodies seem particularly defined and changed by watery flow, Viola's story suggests how the same forces coursing through bodies work outside them, with water activating engines for industry, trade and empire. Drawing on oceanic studies, decolonial feminist theory, the biology of menstruation and early modern imperial thought, I analyse *Twelfth Night* as awash in ideas about the ways globalisation unfolds across 'hungry' seas and inside women's bodies, with water's hydraulics coordinating how, when and why people and empires might come together.

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> 'Philosophy, that Old European form of thought and life, is undeniably exhausted; biosophy has only just begun its work'. (Peter Sloterdijk, *Foams*, 24–25)

> 'Organisms are creative responses, improvisations with and through their capacity to become one with their environments' (Malin Ah-King and Eva Hayward, 'Toxic Sexes').

> 'I am for all waters'. (Feste in *Twelfth Night* 4.2.64)

## Introduction: sex and the hydraulic city

Shipwrecked near Illyrian shores at the start of *Twelfth Night*, Shakespeare's Viola discovers that she's also been violently torn from her brother Sebastian,

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presumably lost at sea. Sebastian's disappearance is devastating, although it is not the first time the twins have been divided, nor is it the first time that liquid forces have pulled brother and sister apart. What's different this time – and what I argue Shakespeare explores in *Twelfth Night* – is the way water provides an instrument to reunite the pair after reorganising their surroundings in Illyria. In a New World primed for new forms of relating, where other beings are revealed to be riven and reconstituted by fluid feelings and desires, water proves to be a solution as well as a solvent. Operating as a force of both renewal and resistance, water reroutes human agendas in a play performed just as the East India Company's new charter has given it a monopoly over the trade of cotton, spices, tea and indigo, enabling it to compete with Dutch and Portuguese traders in the region.[1] These new trading plans underwrite truths which *Twelfth Night* banks upon, that the same things which water splits apart will come back together, improved and recalled with time, revealed as never lost. Liability and misfortune are both shared and limited in this new system and Viola thus gambles that because she has been saved, 'so perchance may [her brother] be' (1.2.5).[2] Operating as both a fact of nature and an instrument of power in this watery world, change simultaneously upholds authority and regeneration, and comedy and empire filter through this liquid infrastructure, too.[3]

'The vastness of the global ocean was perhaps the major conceptual 'discovery' of early modern European navigators and cartographers', according to Dan Brayton, who characterises the sea as a 'cosmic agent' and 'powerfully disruptive force'.[4] '[F]oundational stories of Western culture' tend to suppress or deny what Brayton calls 'humanity's aquatic or aquacentric ontology', even though our body's blood, sweat and tears share their chemistry with seawater, but Brayton points to Shakespeare's work as an important exception, regularly uncovering an 'ontological connection between humans and the sea'. I argue that these connections extend in multiple directions because Shakespeare's thinking about this liquid infrastructure involves ideas about bodily flow, encouraging him to view the human body of water as something which interacts with larger bodies of water outside. In making this claim, I also draw upon Gail Kern Paster's rich work proposing that early modern ecology and psychology overlapped in a theory of bodily humours, a 'conservative biological discourse' which prescribed behavioural norms but also explained deviations in terms of a shared infrastructure shaping temper and temperature.[5] Paster's valuable model might be opened up more fully, however, because even without an understanding of endocrinology or the hormonal basis for bodily growth and development, the early-modern body was construed as an information management system where individual chemistry interacted with the outer environment. Perhaps the 'humoral reprogramming' which Paster says can occur in Shakespeare's plays is even possible because of the way human moods and

proclivities were not derailed by the environment but organised by it, such that when Hamlet talks about opposing a 'sea of troubles' (3.2), he envisions ending his life rather than saving it.

Both resource and symbol, water streams through the environment and pulses in the blood and managing it entails marshalling the forces of the cosmos. Such a command might generate health, wealth and happiness, too. While literary scholar Steven Mentz links modernity to shipwreck because early modern 'stories of maritime disaster provided ways to think through changing views of humanity and the natural world',[6] viewing water as a powerful agent need not require thinking about it as antithetical to human life. Nor must we limit fluidity to something modern or revolutionary, as Zygmunt Bauman proposes.[7] Instead, early modern water tides and currents could be taken for granted, invested in, bet on, capitalised upon.[8] Rather than viewing water as hostile or disruptive, alien or messy, we might instead simply see water as an early-modern mirror, feeding and fed by human experience, both spring and pool, reservoir and unifying force, image of and source for human desire, growth and identity.[9]

At the same time, water has a history, and scholars of oceanic studies like Mentz investigate how English imperialism and trade took shape in the early modern period in terms of controlling the sea,[10] particularly after the 1588 defeat of the Spanish Armada. Commanding water was increasingly as important as amassing territory to early modern imperial thinking, and while historian Corey Ross locates efforts to build a 'liquid empire'[11] in the machinations of nineteenth century nation-states, Ross's descriptions of 'colonial intervention in the hydrosphere' as 'an essential part of the story of modern imperialism' underscore the ways even earlier thinkers like Shakespeare represent civilization as something negotiated with the seas. These negotiations also expanded ideas about nature as something global and the global as something instrumental, converting the ocean into what engineers like Deb Chakra call a 'charismatic megastructure' or intricate complex, which both rationalises and eases the flow of people, wealth and goods.[12]

The transformations delivered by early modern sea-voyaging were just as dramatic. While Ross emphasises how 'human interactions with water are intimately bound up with questions of hierarchy, inequality and power' (6, 10), philosopher Peter Sloterdijk observes that the phenomenon of 'wet globalisation' also meant that 'where losses of form were bemoaned, . . . gains in mobility [were] now in evidence'.[13] That an airy sense of expansion and change can be tied to ideas about order and value is evident in the metaphors of Eastern powers and treasures scattered throughout *Twelfth Night*. Twice we hear about 'Sophy', the shah of Persia (2.5.170, 3.4.269), and one of Sir Toby's terms of endearment for Olivia's housemaid Maria is 'metal of India' (2.5.11-12), her treasure somehow exotic but also readily available.

Later, when Maria likens Malvolio's smiling face to 'the new map with the augmentation of the Indies' (3.2.74), she reduces the faraway to something odd and off-putting, unthreatening and also part of Olivia's household. These various references to the East subtly shift England's borders, rendering outer limits and further shores with extra details in a furrowed image of a world folded back on itself. The entire globe becomes accessible rather than unfathomable, with merely more places of contact, trade and wealth. Sea captains and pirates get stuck on land in Illyria and, rather than a cosmic unmooring, the sea seems to supply a site for exchange, homecoming and recursiveness.[14]

Indeed, with newcomers safely positioned alongside long-time inhabitants in *Twelfth Night*, Illyria is less of an Arcadia or Elysium[15] and more of an outpost in a global setting, allowing Shakespeare to imagine how the world might be transformed with England navigating wider seas. This preview unveils a system which keeps changing, however. By 2050, marine biologists predict that the sea will be 50% plastic,[16] and this disastrous ecological forecast suggests how water's powers to dictate human projects can be dissolved by tendencies to absorb them. As I explore below, with reference to oceanic studies, female body chemistry and early modern imperial thought, Viola's ambitions to recover Sebastian and preserve herself are no less far-reaching or exploitative. The very first lesson she derives from the cosmic power of the sea prompts her to make use of the capacity, fluidity and liminality of the self and water provides her with an excuse as well as a methodology to hide who she really is.[17]

## 'To Be or Not Be a Eunuch'

The sea makes things happen in *Twelfth Night,* but it also unmakes Viola. After being shipwrecked, she rediscovers herself next to other fragile, dynamic and 'miscible' things Sloterdijk describes, like tears and mist and spray (27-29), equally liable to reform or disappear, part of a system which changes as she changes. Yet the female body, Cat Bohannon explains, is already a work-in-progress,[18] always unfolding in and with time and Viola may now merely sense her participation in and contribution to a larger system surrounding her, one that Shakespeare fills with fertile possibilities for new life and easy withdrawal. Actually, alongside the play's unusual array of references to water in terms of 'rain', 'brine', 'foam', 'drowning', 'drink', 'salt waves', 'tempests', 'Noah', 'scurvy', 'gall', 'Lethe', 'floods' and 'hoisting sail', we might add the suggestion of a uterine environment which protects Viola by pulling her back within the safety of the womb, preventing her from drowning even as it seemingly separates her from her brother.[19] Like Botticelli's Venus rising from the depths of the primordial sea, Viola is roughly reborn at the start of the play. But maybe she is even more like an Eve split apart from her Adam in liquid space, rescued only

because she is separated from Sebastian. Illyria's waters also resemble the bath of amniotic fluid, which upholds and divides embryonic life, allocating resources, directing signals and cleaving tissues, water running through and organising the Garden of Eden in *Twelfth Night*.

Deciding to adopt the disguise of a eunuch and call herself Cesario after being rescued, the new birth narrative which Viola fashions both obscures and includes Sebastian, allowing the sister to hide while keeping the idea of her brother alive. It's as if shutting off bodily flow in one place might force it open somewhere else. Viola doesn't merely assume her brother's gender, after all, but abolishes her sexuality in calling herself a eunuch, the costume of Cesario inviting 'some obstruction in the blood' like the one Malvolio's yellow garters will seem to provoke in him (3.4.20), suppressing rather than diverting healthy bodily responses.[20] We now know how hormones and glands coordinate the complex process of puberty and also how environmental factors play a role in the process, but Viola instinctively avails herself of such machinery, pushing forward by pulling backwards, managing the fluid nature of her being by resetting the clock.[21] Shortly after this shift, Malvolio will disparage Cesario's condition by referring to this newcomer as stuck in 'standing water' (1.5.153).[22] Yet being stuck means gaining ground. Figuratively intervening in the body's hydraulic structure, Viola jams up grief so that she can enter the world once more, this time before she needs to split from Sebastian.

Viola's engineering efforts transform the groundwork for who she is while reorganising the space in which her being might unfold, reconfiguring the system that holds people and things together in Illyria. At the start, Viola mourns her twin by making herself disappear, or at least flushes herself away as an autonomous entity, with water's 'restless', 'unreliable', and 'essenceless' nature supplying relief, delay and a guide.[23] If such interventions are biological as well as psychological, Shakespeare indicates that they've long been at work in the twins' relationship; however, when Sebastian later reports that their father died 'that day when Viola from her birth/Had numbered thirteen years' (5.1.238-39). Presumably, a birthday is something which Viola shares with her twin, so Sebastian's account is odd and revealing, linking disaster with fertility and loss with female puberty.[24] '[T]hirteen' marks the age which divides Viola from Sebastian and both of them from their father; but while female adolescence and the onset of menses appear to prompt her father's death, the sea now also suggests to Viola a means of recovery, modelling ebb and flow and linking female bodies with the rhythms and cycles of the moon and tides. Viola isn't starting from scratch, in other words. What moves away might return again and so thinking of herself as a eunuch removes whatever split the twins apart in the first place, with Viola this time controlling how and when she might be 'delivered to the world' (1.2.39).

Twenty-first-century scientists are catching up to Viola's thinking. As they turn their attention to reproductive health and justice, these researchers are paying more attention to menstruation as a global health phenomenon and women's rights issue, as well as an index of pollution, stress, race and poverty – as 'ecological' rather than 'pathological', the way early modern commentators more often understood it. Although early modern writers like Jane Sharp connected 'menstruation' with the 'monstrous',[25] nowadays scientists align menstruation with maintenance. Documenting how female flow is bound up with external factors, they also demonstrate how the operations of human bodies make their way *outside* by exploring new evidence about the '*bidirectional* impact of endocrine disorders and the physical, chemical, biological and social environment of the individual' (*my italics*).[26] Shakespeare may have similarly seen menstruation as keyed to ecology and community in that it aligns women with cosmic forces while dividing them from men,[27] less a source of defilement or disability than evidence of intricate engineering. In *Twelfth Night*, the playwright simultaneously imagines that the flow which distinguishes men from women might also bring them back together. Such a perspective about hysteria and humours and (what we now see in terms of) hormones permits the madness Shakespeare's Lear evinces and 'antic' behaviour Hamlet displays to appear as manageable, even predictable, especially when water is recognised as the basis for human emotions and environments, its flow appearing natural rather than threatening.[28] Maybe the only thing really new about modern conceptions of an endocrine system stimulating physiological processes surrounding metabolism, fertility and weight is the added presence of chemicals and neurological pathways operating in the same watery infrastructure that organises the rest of the world.

So different is the physiology of Hamlet's grief in a play probably performed at the same time as *Twelfth Night*, roughly 1600-1601. In both works, Shakespeare represents duelling, forged letters, the threat and charm of child actors,[29] rough bundlings of funerals and weddings and drunkenness; pirates fleetingly show up in both plays as well.[30] One wonders what would've happened if the Danish Prince had been shipwrecked on Illyrian shores, given the ways both plays explore the distress experienced by a young person after the devastating loss of a cherished family member. *What if comedy is tragedy plus water?* Seeing herself as a eunuch allows Viola to intercept loss from inside herself, altering history by reconfiguring her body. But Viola's fashioning of herself as a eunuch also attempts to rescue her from hormonal change, returning her to a period before she gets her period.[31] Hamlet is unable to fix the clock even though he perceives the time as broken and disordered (1.5.190-91), yet Viola can adapt her own body to make these adjustments and pilot Illyria's waterways, even if her 'pipe', as Orsino notes, is 'small' (1.4.32).[32] Using comedy to substitute

female pain for male pain, maybe *Twelfth Night* rewrites *Hamlet* by deploying female bodies as more effective conveyors of affect and loss and better filters of grief, desire and power.[33]

Perhaps this is why Viola's rescue and rebirth – unlike those represented by the stories of Robinson Crusoe or Shakespeare's Caliban – also involve improving conditions for Illyria's other inhabitants: *she herself is a pipe*, a tool which not only links inside to outside but also establishes routes to other bodies. Making herself a eunuch jams the flow, but Viola's agency has ripple effects, because seeing herself as part of a larger system enables this system to function more smoothly. Undertaking the reproductive labour which changes the flow of things around her, we might see Viola as part of the material environment, as integral as any bridge or tunnel, her machinations as necessary as those of some switch or valve. In bringing water ashore with her, Viola not only modifies how the world sees her but also challenges that world's structure. Female colonists who later arrive on other shores will proceed almost as skillfully. After a kidnapped maiden is shipwrecked in Margaret Cavendish's *The Blazing World* (1666), for instance, she becomes an Empress who eventually presides over the globe and stages her imperial power on the sea. The Empress wages war on the rest of the world when her homeland is attacked, making the ocean both the seat of her power and global vanishing point, both glue and solvent at once.[34] Imperial space is amorphous, watery and planetary in *The Blazing World* and the Empress returns as both saviour and terrorist, 'appear[ing] to walk upon the face of the water'.

While water scrambles geography in Cavendish's romance and cancels renewal in a dramatic picture of absolutist tyranny, it introduces even more disorder in *Twelfth Night,* so that there's no path back home or impulse towards returning – or even any place beyond Illyrian waters; actually, everyone ends up in Illyria sooner or later. Yet Viola works *with* water rather than *on top* of it, employing it as a tool to heal, rejuvenate, unclog and cultivate, as a resource rather than a weapon, which seeps into the Illyrian landscape. Viola's efforts are outlined at a time when the East India Company was beginning to establish trading posts and ports in Madras, Bombay and Calcutta, converting the Indian Ocean into a site of expansion, trade and renewal. Shakespeare illustrates the effects of this transformation of the early modern waterscape with his ongoing stream of references to 'tears', 'urine', and 'ale'. Even rain, Feste kindly reassures us, 'raineth every day' (5.1.383), nature cooperating in a process whereby water is converted into a good which stimulates the production of other goods. Rather than being a tool of domination like the one Cavendish's Empress wields, water reorders and stabilises; instead of arriving suddenly or menacingly in the shape of sea monsters or terrifying tides, Illyrian waters are easily supplied and regularly reconstituted by human meanings.[35] Viola simply pours this

New World through her body, restarting how things grow, how feelings are felt and expressed and how people will come together.

## 'Motions of the Liver'

In underscoring the continued importance of the early modern doctrine of the four humours (blood, phlegm, yellow and black bile), Paster offers a materialist explanation for how mental and emotional states could be related to the natural world in Shakespeare's time, produced and modified by everyday interactions with the environment, particularly heat and cold.[36] Katharine Park similarly describes an early modern 'hydraulic' system connecting psychological and physiological responses, picturing human bodies as constantly working to relieve pressure from within.[37] These interactions are mostly unsettling and divisive, however. Messily spilling or leaking fluids, Shakespeare's comedies, according to Paster (2003), 'explicitly invoke' 'the language of the humours'' such that 'casual, subthematic locutions of bodily affect and experience seem to organise that which is being thought and said' (48). This humoral language also distinguishes characters from each other – Shylock's obstinacy attributed to one kind of cocktail of humours, Falstaff's corpulence to another (2003: 50-51).

By ignoring the primacy of water in the mix, these formulations leave too much of the world outside, however. Maybe this is why the imbalance which Paster and Park inevitably find is internal rather than external, with separate bodily environments categorising people through proclivities and temperaments, class and rank, rather than reinforcing their connections to the larger world. Yet water brings people together as well because culture also operates under the skin, rendering biology a *local* phenomenon which cannot be abstracted, a product of the environs through which water runs and surrounds.[38] This means that if water carries and helps constitute the humours which stream through what Paster calls the 'semi-permeable, irrigated container' which all bodies provide, water then presses outward, too, spilling into other bodies, soaking other settings.[39] Even as the meaning of 'humour' changed over time, dematerialised and transformed into wit or mood,[40] water remains behind, spilling even now, pouring in and out of incontinent, oceanic bodies which continually swell, grow and leak.

This also suggests why, after Viola adjusts local biology by tinkering with her monthly flow, those changes trigger other environmental disturbances in the rest of her world, imbuing and diverting Illyrian passages and transforming its infrastructure of emotions, bodies and relationships. We see this right away in Cesario's first interactions with Olivia. Scientist Juliet Livingston maintains that 'historical transformations' 'produce new biological and sociocultural norms' which reorganise the body and stir the imagination,[41] a shift we perceive when the grieving Olivia meets Cesario and is immediately encouraged to experience feeling as something pleasurable and view

her body as a conduit for desire, curiosity and wit. The 'hydraulic system' Park mentions has branches inside other bodies, in other words and, as any other infrastructure, the waterworks Cesario introduces conveniently expand. Actually, Cesario's petitions to Olivia have almost instantaneous physiological effects, with alterations quickly registered inside and outside Olivia's body. Sloterdijk claims that 'biosophy' rather than 'philosophy' more aptly envisions a system which puts bodies, power and thought into dynamic relation with the world and his 'philosophical gynaecology' explains how human bodies enter the world already conjoined with it, co-existent 'with the other, the outer'.[42] Acting as midwife, Cesario helps Olivia not only to see her place in a larger system but also talk about it in a way that acknowledges she is both figure and ground, *of* and *in* the flow. 'Fate, show thy force, ourselves we do not owe./ What is decreed must be', Olivia announces once Cesario departs, finding a way to connect their two fates and two pipes. Inserting herself into this system's workings as both agent and outcome, she uses poetry to pull things together, concluding with a rhyme 'be this so' (1.5.298-301). Cesario's visit has 'deliver[ed]' Olivia to the world, too, allowing Olivia to reconnect her body to her mind and her feelings to other people.

At the start of *Twelfth Night*, Shakespeare's audience had been told that a grief-stricken Olivia had vowed to cloister herself for seven years. Locked in her chamber, this space was slowly filling with her tears or 'eye-offending brine' (1.1.27-30); removed from the world, Olivia's beauty will fade and wither. The undaunted Duke Orsino believes that his love, 'as hungry as the sea' (2.4.99), can rescue his beloved by flooding Olivia's 'dirty lands' (2.4.80), but this plan to engulf Olivia only repels her. Shakespeare's Sonnet 135 plays with a similar scenario by envisioning a beloved whose inability to refuse an eager lover explains her appeal: 'Wilt thou, whose will is large and spacious,/Not once vouchsafe to hide my will in thine?' The reasoning behind the male's plea to his female beloved is simple: 'The sea, all water, yet receives rain still,/And in abundance addeth to his store' (ll. 6-5, 9-10). Yet although this sonnet describes the female body like the sea, welcoming and unconscious, the female (unlike the pointedly coded masculine sea) cannot properly husband her resources or manage 'abundance'. Orsino presumes he can easily drain this ocean, telling Cesario that there's 'no woman's heart/So big, to hold so much. They lack retention./ Alas, their love may be called appetite,/No motion of the liver' (2.4.95-97). Orsino and Shakespeare's sonneteer share a picture of the female body as inert and depleted, indifferent rather than expansive. Cesario's interactions with Viola, in contrast, imagine the female body as a crucial resource feeding the sea as well. Rupturing Olivia's 'presumptuous' sense of 'bodily autonomy', as Paster (1999) puts it, Cesario reminds Olivia about her necessary place in this network: 'What is yours to bestow is not yours to reserve'.[43]

The exchange of wit in their conversation results in new circuits of feeling through which other ideas and goods – including rings and formal expressions of desire – can now safely travel and be readily recalled, establishing a feedback loop which even rewards feelings of pain.

Certainly Olivia is apprehensive about the strange flood of feeling she experiences upon encountering Cesario: water might be an instrument of disease and feelings might spread like a virus or erupt like something contagious: 'How now?' Olivia wonders: 'Even so quickly may one catch the plague?/Methinks I feel this youth's perfections/With an invisible and subtle stealth/To creep in at my eyes' (1.5.285-88). But water can push in multiple directions and just as Olivia opens herself up for feeling, she simultaneously construes others as equally receptive. Wittily 'inventor[ying]' her lovely features in 'divers schedules' of her beauty, she makes herself available for immediate inspection and circulation (1.5.234), recognising her body as operating in concert with and attuned to other bodies. In the new order Viola sets in motion, emotions are construed as physiological reactions to the outer and other, such that feelings must flow and 'hearts', as Olivia likewise comprehends, 'can move' (3.1.61).

Moreover, Olivia starts to grasp how her body is shaped in and by time, influenced by an ecology which not only joins her with Cesario but parts them as well. The next time the two meet, Olivia explains how their relationship is coordinated: 'The clock upbraids me with the waste of time./Be not afraid, good youth, I will not have you;/ And yet when wit and youth is come to harvest/Your wife is like to reap a proper man' (3.1.128-131). Admitting the possibility that Cesario cannot love her, Olivia confesses that there are circumstances where she might instead be moved by Orsino's love (3.1. 121-22, 161-62). But she observes that Cesario will change, too and that such transformations will happen along with and within seasonal shifts. What's inside bodies makes them *flow together* and this transformation of Olivia's thinking and feeling occurs because she learns to think of herself as a pipe, too. Even in her first meeting with Cesario, Olivia had started to grapple with her body's engineering: 'Not too fast' (1.5.283), she pauses herself, double-checking her reactions. Even here, she involves others in these biological processes, admonishing Cesario that ''Tis not that time of moon with me to make one in so skipping a dialogue' (1.5.192-93). If Olivia alludes to her own menstrual cycle in this scene, she also indicates that the tidal motions which affect her are 'conjoined' and 'co-existent' with those people nearby.

Sir Toby can't resist trashing this picture of ecological order and natural flow and Shakespeare's audience hears him joking about '[m]ak[ing] water' 'in a cinquepace', punning about peeing in a garbage dump (1.3.122-27).[44] Malvolio's attentions to Olivia are similarly crass and reductive, concluding that the letter Maria has forged comes from Olivia's hand because it displays

Olivia's 'great P's' (2.84). Both men reduce the flow to urine, fluid waste that is unhealthy and polluting. Almost like the ring Malvolio carelessly tosses to the ground during his awkward conversation with Cesario, water is seen both by him and Sir Toby as inactive and worthless – much like the female appetite Orsino describes, something 'lack[ing] retention', too. Malvolio will mistakenly report that Cesario is the one who 'peevishly' 'threw' the ring away, but Cesario instead retrieves the ring, taking it as a sign of Olivia's susceptibility to a 'pregnant enemy', all too easily activated by the machinations of another body (2.2.27). If male and female bodies switch places in Cesario's formulation and even get tangled by Viola's hydrology, the dangers are clearly tied to growth and fertility and chances for renewal, with a 'pregnant enemy' exploiting and exploited by the very same vulnerabilities. Time must 'untangle' the knot, but not, as Viola tells Shakespeare's audience, sever it.

Because of the way she expands and reroutes Illyrian waterways, we might therefore see Viola as destabilising flow in still more dramatic ways, just as the bogus letter Maria fashions provides another way to jam the system, submerging her own desire to climb the social ladder by exposing Malvolio's secret fantasies to do so. Some contemporary scientists and environmentalists maintain that similar disturbances to flow are caused by endocrine-disrupting chemicals (EDCs), pollutants or irritants which upset an ecosystem, as when medicines that are excreted re-enter our bodies through the drinking water. What disrupts Viola ramifies through the rest of Illyria,[45] and the changes she introduces when she hacks into her system are not entirely under her control. Data on EDCs collected by the World Health Organisation links such chemicals to early puberty, increased cancer risk and birth defects, but scholars like Malin Ah-King and Eva Hayward downplay those risks, suggesting that all life is disruptive, anyway: our bodies, 'flush with chemicals', 'become themselves in chemical relations'.[46] Shakespeare is more ambivalent about the outcomes in *Twelfth Night*, mostly broadly calculating the scope of the changes which Viola gets underway, with less attention to the details of later outcomes and hasty couplings. '[M]otions of the liver', whether slight or great, are necessarily a part of a larger geography, however, and like other local disruptions introduced by early-modern colonialism, trade and sea travel, Viola's adjustments to bodies and futures have 'multiple connections that spread out on vertical and horizontal scales'. Time, space, bodies, societies, wills, marriages, inheritances and reunions all get reshuffled in the play as a result.

'[C]ontravening order at one scale and reproducing it on another',[47] Viola's altered experience of flow becomes the medium through which nearly every other desire in Illyria now travels, her body establishing new conditions for the other exchanges enacted in Illyria. When Sebastian finally reappears, he is immediately married to Olivia and a 'reprogrammed' Olivia doesn't mind the trade-off; Orsino is equally primed to shift his

attentions to Viola, with or without her disguise. We can include the midsummer madness of Sir Toby and Maria's marriage here as well. As if some switch has magically been turned on, human ties have been comprehensively augmented and pressed out in other directions, bodily chemistries and temperaments prompted to shift gears with just the right signal, too. To be sure, the conventions or mechanisms undergirding Shakespeare's comedy ensure these smooth transitions and happy endings, but water's ripple effects inspire such results, too, repairing a broken world primarily by creating new pathways for love, rebirth and other social goods. Commenting on the way Viola opens up new channels for hope in *Twelfth Night*, Rikita Tyson cites Alexander Leggatt's claim that the abrupt resolution of *Twelfth Night*'s love plot is enabled by an 'impersonal, organic process', yet Leggatt's reading overlooks the larger machinery Viola has set in motion, pulling Orsino and Olivia downstream with her.[48] Revising her past allows Viola to overhaul Illyria's waterworks, making it possible to introduce other changes which seem poised, in turn, to reshape the rest of the world.

### '*The Spinsters and the Knitters in the Sun*'

Indeed, we might even suggest that Viola's humble DIY engineering project aligns with imperial efforts to develop the world's resources by controlling watery flow. Although scholars of oceanic studies emphasize male travel, male bravery and male settlement in these efforts – like Feste, who explains Orsino's lovesickness in terms of wanderlust and says 'I would have men of such constancy put to sea, that their business might be everything and their intent everywhere' (2.4.74-76) – we can speculate that Viola's reconfiguration will generate more business overseas. London's merchants relied on the efforts of poor women to produce the woollens and other English textiles increasingly needed to trade for Indian spices, cotton, and indigo, and Feste provides some details about these punishing labours and the scores of unmarried women undertaking them in Act 2, shortly after Cesario's first visit to Olivia. The 'spinsters and the knitters' and 'free maids' who provide the source for the love song Orsino wants to hear in this scene synchronise their labors by 'chant[ing]' as they 'weave their thread with bones' (2.4.41-45), coordinating their bodies with the needs of the global market. While water, like air and ice, resists colonisation, early modern women's bodies proved unusually serviceable,[49] and although unchaperoned like Olivia and Viola, the women Feste describes are also invisible and typically anonymous. Producing lace and spinning cloth, their difficult work keeps them alive while enriching England's coffers at home and expanding trade abroad.

Shakespeare establishes both the value and the dreariness of such work in Act 2, emphasising the women's numbers to make them a crucial if unseen[50] key to England's growth as a 'liquid empire'.[51] These labors also supply a means to remove single women from the larger society – or,

perhaps, reposition them: what had formerly been a drain now becomes a pool when poor women are taken out of circulation and removed from Illyria's marriage market (as they were from England's marriage market) with their reproductive powers siphoned off so that they might instead produce cheap items for export overseas. Female fertility is jammed and diverted just as it is when Viola makes herself a eunuch, but in this case, the aims are commercial, and the flow of money and goods can be pushed across the seas. The song Orsino requests describes the misery of a man 'slain by a cruel maid' who needs his 'shroud' 'prepare[d]' (2.4.54-55), but this music also ensures that poetry rather than protest emerges out of female collaboration, all the while guaranteeing women's safety and industry. Such a hydraulic system also converts one kind of pain into another kind, processing male tears through female laments while using water to widen the flow of English gold and silver alongside Indian cotton, spices, indigo and tea. We might even conclude that Shakespeare's play becomes part of this global infrastructure, too, involving itself in the hydraulics through which the pain of women too miserable to deserve notice is transformed into wares which rich men like Orsino can traffic in.

The early modern ballad 'A warning for Married Women' (EBBA 31991) occupies a similar place in this early modern system linking water, women and wealth, in this case issuing a 'West-Country Woman' a summons to the sea where abundant riches and love await her. What's especially striking about this ballad is the ease with which a married woman's household might dissolve: like Shakespeare's shipwrecked Viola and Sebastian, another happy family must come undone for globalisation to work its magic. The ballad begins with Jane Renalds's promise to marry James Harris, a 'Sea-man brave', and she's prepared to wait when James is suddenly 'prest to Sea', gambling like Viola upon her beloved's return. Yet when she receives news of James' death 'within a Foreign Land', Jane marries a Carpenter who gives her 'Three pretty Children'. Years later, however, a 'Spirit in the night' calling itself by her first lover's name lures Jane to sea with a vision of 'Mariners and Merchandise' 'at [her] command'. Invisibility, money and escape are the magnets as well as the rewards in this scenario and Jane seems eager to leave her old life behind. The ballad's 'Warning' is *for* 'Married Women', not *about* them, however, because water is an instrument of losses and gains occasioned through and by women's bodies. Jane is more of a victim than a villain in this story. Although James Harris's Spirit possesses 'seven Ships upon the Sea', the couple never make their way back to England or to Jane's children. Instead, she becomes part of the circuit of goods flowing elsewhere, her body another pipe through which English wealth might grow; removed from home and severed from her children, Jane seems destined to remain at Sea, unable to return the way James Harris can.[52]

## Conclusion

Jamming female flow the way Viola does leads to comparable rewards for Illyrian society and East India Company revenues only increase as this process further expands. We might even envision Olivia's 'dirty lands' as inhabited by unmarried women unavailable for exchange but ready to work,[53] their carefully coordinated labours simultaneously driving the flow of goods and wealth across the seas. Once new spaces are opened up for communication, transaction and receptivity, new markets, appetites and desires can be stimulated as well and empires made to work *inside* bodies, too.[54] This coordination unfolds on a relatively small scale in *Twelfth Night* where both Viola and Olivia mourn at the same time and then become ready for courtship simultaneously. If these women's fears or reservations have been revealed as 'bugs' in the system, evidence of bodies in need of recovery and repair, Shakespeare's play also helps us imagine the effects when other women's bodies can be similarly managed (or updated or streamlined, maximised or optimised) by an even larger watery system. Such a global enterprise governs flow by bringing populations under control and deciding when and which women can mature, while ensuring that we see this liquid organisation of female bodies as essential and inevitable – even ideal, harmonious and generative – as a fact, that is, of nature.

## Notes

1. Marion A. Taylor explores the reference to the pirate ship which *Twelfth Night*'s Antonio takes command of (5.1.63) in 'He That Did the Tiger Board.' *Shakespeare Quarterly* 15, 1 (1964): 110–113. Taylor notes that contemporary accounts mention a ship named the *Tyger* conveying Queen Elizabeth's emissaries and traders which had recently requested safe passage in Indian waters.
2. Rikita Tyson explores Viola's reasoning at the start of the play, construing her 'being' as something both negotiated and willed, deliberately formed and easily dissolved, shaped by both 'volition' and 'submission.' See 'What They Will: Comic Grammar in *Twelfth Night*.' *Shakespeare* 10 (2014): 158–70 (160-61). Like Tyson, I see Viola's time in Illyria as a mechanism for splitting her completely from her twin. Randall Martin makes a different case for the twins' continued sharing, claiming that 'Shipwreck and foreign survival traumatise Viola and Sebastian and permanently dispossess them of their former lives.' Still, Martin's treatment offers a valuable account of how the 'aftershocks' of the disaster structure the unfolding of Shakespeare's play, shape radically new subjectivities for brother and sister, and disturb the festivities associated with Twelfth Night, its 'carnival rhythms of release and restraint.' See 'Shipwreck and the Hermeneutics of Transience in *Twelfth Night*.' *Twelfth Night: A Critical Reader*. Eds. Liz Oakley-Brown and Alison Findlay (NY: Bloomsbury, 2014): 123–43 (123-25).
3. Many scholars have helped me think about the watery infrastructures of the early modern body and early modern imperialism, and I am especially

indebted to my colleagues Rob Higney for pointing me in this direction in the first place, and Doris Barkin, for some initial reactions. I also want to acknoweldge the generous support of The Rifkind Center for Humanities and Arts at the City College of New York. Rebecca Oh shrewdly analyses genre as 'the infrastructure of infrastructure' assembling different elements of literary texts and coordinating our expectations about how parts contribute to wholes in 'How Things Will Go: Genre, Infrastructure, and Hope in *Welcome to Lagos*.' *Ariel: A Review of International English Literature* 54, 3–4 (2023): 11-36. Oh emphasises comedy's impulse towards hope and possibility while I focus instead on the comic genre's ambitions for social order and conformity, but I am enormously indebted to Oh's work. See too Brian Larkin, 'The Politics and Poetics of Infrastructure.' *Annual Review of Anthropology* 42 (2013):327-43 (see esp 329-331, 337).

4. See Dan Brayton, *Shakespeare's Ocean: A Ecocritical Exploration* (Charlottesville: U Virginia Press, 2012) 2; see also 7-8, 137, 43.
5. Gail Kern Paster links the body's 'weather' with Illyria's social climate in 'The Humor of It:Bodies, Fluids, and Social Discipline in Shakespearean Comedy.' *A Companion to Shakespeare's Works*. Volume III. Eds. Richard Dutton and Jean Howard. NY: Blackwell, 2003: 47-66.
6. See Steve Mentz, *Shipwreck Modernity: Ecologies of Globalization, 1550–1719* (Minneapolis:University of Minnesota Press, 2015) xxvi. Commenting on the 'ubiquity of sea imagery during the sixteenth- and early seventeenth centuries,' Susan Rojas explores how Shakespeare imagines maritime disaster in 'Shipwreck, Immersion, and a Great Sea of Joys: Watery Tropes, the Blue Humanities, and Affective Ecologies in *Pericles*.' *Early Modern Studies Journal* 9 (2023): 50–68 (52). See, too, Julie Sanders, 'Water Memory and the Art of Preserving: Shakespearean Comedy and Early Modern Cultures of Remembrance,' Heather Hirschfeld, ed., *The Oxford Handbook of Shakespearean Comedy*, Oxford Handbooks (2018; online edn, Oxford Academic, 9 Oct. 2018), https://doi.org/10.1093/oxfordhb/9780198727682.013.12, accessed 9 July 2025. Sanders's focus on saltwater as preservative in *Twelfth Night* emphasises Olivia's commemorative efforts over Viola's labours to reorganise and repair herself and others in Illyria.
7. Zygmunt Bauman views water as a revisionary, even revolutionary, but mostly figurative force in *Liquid Modernity* (Boston, Polity Press, 2000), where he offers a picture of melted structures and fluid meanings. For a discussion of water as a traditional resource with sacred, therapeutic, cleansing, and purifying powers, see Veronica Strang, *The Meaning of Water* (NY: Routledge, 2004). For Strang, thinking about water automatically means thinking about bodies and order rather than about disorder and disruption. See, too, *Liquid Empire: Water and Power in the Colonial World* (Princeton: Princeton UP, 2024), where Corey Ross observes that even monsoons can be seen as regularly-occurring seasonal disruptions.
8. Mentz emphasises the risks of sea travel in 'Toward a Blue Cultural Studies: The Sea, Maritime Culture, and Early Modern English Literature.' *Literature Compass* 6,5 (2009): 997–1013, arguing that in the early modern period relatively easy coasting in the Mediterranean basin

   was replaced by longer and more treacherous voyages around the horns of Africa and South America (1000). See, too, Mentz, *At the Bottom of Shakespeare's Ocean* (NY: Bloomsbury, 2009) where he claims that *Twelfth Night*

dives into a 'pre-history' of understandings of water's 'destabilising force' (51, 54). For Brayton, the sea is accessible, navigable *liminal* space (143), while Mentz repeatedly focuses on the sea as dangerous, forbidding, inhuman *other* space ([2015] xxix; 'Blue Cultural Studies,' 1002).

9. See Mentz, 'Blue Cultural Studies' (998).
10. See Paster (2003) 56-8. For discussions of the ocean's changeability, see Brayton 2; and Laurence Publicover, 'Pericles's Humming Waters: Nonhuman Agency, Textual Criticism, and the Practice of Material Ecocriticism.' *Shakespeare Quarterly* 73, 3–4 (2022): 280-302. Hester Blum provides a useful overview in 'The Prospect of Oceanic Studies.' *PMLA* 125, 3 (2010): 670-77.
11. See Ross, 6-10. For a more detailed account of how the British empire transformed India's water into a commodity to which only select citizens had rights, see Nikhil Anand in *Hydraulic City: Water and the Infrastructures of Citizenship in Mumbai* (Durham: Duke UP, 2017). For related discussions of England and its Eastern empires, see Jyotsna Singh, ed. *A Companion to the Global Renaissance: Literature and Culture in the Era of Expansion, 1550-1750*. (NY: Wiley-Blackwell, 2021).
12. Deb Chachra, *How Infrastructure Works: Inside the Systems that Shape Our World* (NY: Riverhead Books, 2023): 2-5.
13. Peter Sloterdjik, *Foams. Spheres III. Plural Spherology*. Trans. Wieland Hoban (South Pasadana, CA: Semiotext€, 2017) 25. In *Shipwreck Modernity*, Mentz (2009) cites Sloterdjik as concerned with the 'oceanic nature of European globalisation' (xx). But where Sloterdjik's attention to foams emphasises evanescence and ambiguity at the centre of the modern imaginary, Mentz cites rupture and disaster.
14. All of my references to Shakespeare's *Twelfth Night* make use of the Oxford World Classics edition prepared by Roger Warren and Stanley Wells (NY: Oxford UP, 1994, 1998).
15. See Tobias Döring, 'All the World's the Sea: Shakespearean Passages.' *Shakespeare Jahrbüch* 148 (2019):11-30 (17–21).
16. This statistic has been reported widely, but my source is the Center for Biological Diversity https://www.biologicaldiversity.org/campaigns/ocean_plastics/ Downloaded 9 August 2024. See also https://theoceancleanup.com/?gad_source=1&gclid=Cj0KCQjw3vO3BhCqARIsAEWblcBMuxYXX5HwHNigdsolrytfhoyJM4kQ_pRK-Y7r6ptW_-Qe7vHOo_QaAi6DEALw_wcB and https://www.pewtrusts.org/en/trend/archive/winter-2022/our-ocean-is-choking-on-plastic-but-its-a-problem-we-can-solve
17. See Mentz, *At the Bottom of Shakespeare's Ocean* (61).
18. Cat Bohannon, *Eve: How the Female Body Drove 200 Million Years of Human Evolution* (NY: Knopf, 2023).
19. Mentz, *At the Bottom*, 50.
20. Such 'obstructions' were more typically treated, according to Patricia Crawford, with bloodletting or vomiting. See 'Attitudes to Menstruation in Seventeenth-Century England.' *Past & Present* 91, 1 (1981): 47–73 (54). Crawford also investigates how menstruation's monthly 'purgations' of 'excrement' or 'Unprofitable Blood' were seen to explain female inferiority (70, 49-50). According to Martin, the disguise of eunuch offers Viola a role which she 'defines (for obvious reasons) not as sexual lack but positively as the multilateral connectivity of courtly performance' (130). See also Keir Elam, 'The Fertile Eunuch: *Twelfth Night*, Early-Modern Intercourse, and the Fruits of

Castration.' *Shakespeare Quarterly* 47, 1 (1996): 1 -36 (1). Elam argues that Viola's disguise and later mention of a dead sister who pined away with lovesickness are both designed to emphasise female continence rather than *incontinence*--or the depth or reach or possible transferability of affection which I explore here. Cf. Mentz, *Bottom*, who claims that Viola's disguise underlines Illyria's Turkishness (54). Emphasising watery difference and disorder, Mentz focuses on Illyria's shorelines rather than its seas, claiming that Shakespeare's play shows us 'how little it takes to unmoor us from the world' (56).

21. See Tyson, 160–61.
22. See W. Roy Mackenzie, 'Standing Water.' *Modern Language Notes* 41, 4 (1926): 283-93. Mackenzie comments that Shakespeare uses 'standing water' to suggest something 'stagnant' but also located in '*both* flow and ebb' (287).
23. I draw on Sloterdjik's phrasing 27–29, 48.
24. Cf. Diane Hunter, who concentrates on the twins' loss of their mother and argues that Viola's assumed name 'Cesario' 'suggests a form of birth, Caesarean, as though the shipwreck dissevering her from Sebastian were a replay of their separation from each other and their mother at birth.' See 'Shakespeare's Continuity Through the Daughter.' *Literature and Psychology* 48, 3 (2002): 38–55 (50-51). Crawford claims early modern girls menstruated when they were 16 or 17, although evidence is hard to come by, and she cites Francis Bacon's mention of the growth of 'secret hair' on girls at 12 or 14 (67).
25. See Crawford, 70–73.
26. See Sanjay Kaira and Nitin Kapoor, 'Environmental Endocrinology: An Expanding Horizon' in National Library of Medicine (2021) https://pubmed.ncbi.nlm.nih.gov/33734659/; Top of Form and Hoi, Jin-Ho and Yoo, Han-Wook, 'Control of puberty: genetics, endocrinology, and environment.' *Current Opinion in Endocrinology & Diabetes and Obesity* 20, 1 (2013): 62-68. Hoi and Yoo consider the impact of modern pesticides, forest fires, radiation, and fast food on the developing bodies of young women.
27. See 'Of Mice and Menses.' *The Economist* 1 June 2024: 64-65, where menstruation is explored as a miraculous and meticulous mechanism 'amounting to repeated injury and repair of the whole lining of the womb' 'like a jigsaw . . . . involving moving parts coming together with exquisite timing.'
28. Although Aviva Farkas concentrates on how women are particularly affected by heat in 'Her Own Humor: Contrasting Philosophical Discourses of Femininity in Shakespeare's Comedies.' *Comitatus* 53 (2022): 165-204, she only considers *The Winter's Tale* and *The Taming of the Shrew* as illustrating some interest in the humours, and only appears to believe that women suffer from humoral imbalances.
29. Hamlet's worry about the fate of the popular child actors threatening to supplant companies of adult performers (2.2.368-69)—'Who maintains 'em/How are they escot'd?' and what will happen when their voices change?—finds relief in the watery solution prompting Viola's disguise as a eunuch: castration, after all, would keep these boy actors permanently safe.
30. Döring proposes that *Twelfth Night* is a 'twin play' which stages much of the same issues of 'mourning, memory, and melancholia' (17-19). Hunter similarly describes *Twelfth Night* as a 'manic counterpart to the depressive *Hamlet*' (40) and proposes that *Twelfth Night* allows Shakespeare to imagine the surviving twin as a son instead of a daughter, repairing the loss of his son Hamnet who had died years earlier and left behind a surviving twin sister.

31. Suzanne Penuel focuses instead on Sebastian's disappearance as part of a 'wistful vision of escape from the constraints of the early modern family.' See 'Missing Fathers: *Twelfth Night* and the Reformation of Mourning.' *Studies in Philology* 107, 1 (2010): 74–96 (90).
32. Gaywyn Moore takes a less sanguine view of women and water in ''Was My Sister Drowned?': Voyaging While Female in *Twelfth Night*.' *Mediterranean Studies* 26, 2 (2018): 159-78. In 'The Chronotopes of the Sea,' Margaret Cohen explores how rescue narratives like the story of Robinson Crusoe typically set the historical clock back to zero in Franco Moretti, ed. *The Novel, Vol. 2. Forms and Themes.* (Princeton: Princeton UP, 2006): 647–66 (659).
33. For an overview of global health explorations of menstruation, see Marni Sommer, 'Menarche.' *Global Health Matters* (2013): 399-401. Later literary treatments of young women's development tend to represent female flow as an unwelcome marker of change and growth. A defiant Jane Eyre rages in a red room at the start of Charlotte Brontë's novel, and 20th century treatments in Jamaica Kincaid's *Annie John* (1985) and (to a lesser degree) Toni Morrison's *The Bluest Eye* (1970) describe adolescent hormonal forces as surprising, embarrassing, even warlike, even if their presence is an interruption rather than something ongoing, a one-time event like a wave which strangely crashes only once. Stephen King's 1974 *Carrie* is a splashy exception where menstruation's return is horrific and destructive, making female fertility coincide with Armageddon. Rolled out the same year as King's novel, Roman Polanski's *Chinatown* links get-rich schemes in California with a colossal water grab; flow must be measured by powerful fathers in this desert setting which attaches the diverting of water to incest and sexual abuse. *Perhaps tragedy is comedy plus drought* – or comedy with male pain. Indeed, Hollywood tends to handle male flow as typically damaged and defeating. See for instance, 'A River Runs Through It' (dir. Robert Redford, 1992), or the 2018 remake of 'A Star is Born' (dir. Bradley Cooper) where the drunken male protagonist urinates on himself in public. Jane Austen's Mr. Elliott alludes to similar male damage in describing sodden seamen in *Persuasion*, where he comments that 'a sailor grows old sooner than any other man' (49).
34. See Margaret Cavendish, *The Blazing World and Other Writings*. Ed. Kate Lilley (NY: Penguin, 1994) 209-210. Dyani Johns Taff explores the Empress's control of the maritime environment, starting with her kidnapping, in 'Precarious Travail, Gender, and Narration in Shakespeare's *Pericles, Prince of Tyre* and Margaret Cavendish's *The Blazing World.' Travel and Travail: Early Modern Women, English Drama, and the Wider World.* Eds. Patricia Akhimie and Bernadette Andrea (Lincoln: U Nebraska Press, 2019): 273–91 (see esp 279-80).
35. For a detailed discussion of water's quotidian nature, see Anand, who proposes that '[d]ifferent waters' are 'everywhere' in Mumbai, 'a city that is surrounded by the sea' (23).
36. Paster, *The Body Embarrassed: Drama and the Disciplines of Shame in Early Modern England* (Ithaca: Cornell UP, 1993): 19.
37. Katharine Park, 'The Organic Soul.' *The Cambridge History of Renaissance Philosophy*. Eds. C. B. Schmitt, et al. (NY: Cambridge UP, 1988): 464–84 (469). Paster (2003) cites Park 48.
38. See Juliet Livingston, 'Pregnant Children and Half-Dead Adults: Modern Living and the Quickening Life Cycle in Botswana.' *Bulletin of the History of*

*Medicine* 77, 1 (2003): 133–62 (135). Livington explores alterations in colonial settings resulting from seeming improvements in standards of living, 'including an earlier onset for puberty in girls and an increased incidence of stroke among older women, thus refashioning the female life course' (133).

39. Paster (1993): 8-11; see also Paster (2003): 54.
40. See Paster (2003): 47-50.
41. Livingston 133. For a related account of the way modern science itself often furthers the goals of empire, see David Arnold, ed. *Imperial Medicine and Indigenous Societies* (Manchester: Manchester UP, 1988). See also Larkin 339.
42. Sloterdjik, 15, 31. Sloterdjik's philosophical project to reconcile bodies with their environments can be contrasted with Amy Koerber's critique of Western ambitions to manage and systematize bodies, especially those of women in *From Hysteria to Hormones: A Rhetorical History* (Penn State UP, 2018), esp. 88-90.
43. See Paster 1999: 34. The index in *The Body Embarrassed* makes no mention of 'menstruation' or 'fertility.'
44. See Paster's (1999) discussion of Sir Toby's joke (30-32).
45. I draw on phrasing supplied by Susan Leigh Star, 'The Ethnography of Infrastructure.' *American Behavioral Scientist*, 43, (3) (1999): 377–391 (387). https://doi.org/10.1177/00027649921955326. In the most egregious cases of endocrine disruption, empires damage one kind of flow to ensure the development of new waterways. Tehila Sasson explores, for instance, how Nestlé persuaded mothers in poorer countries to relinquish breast-feeding for powdered formula, all the while knowing that breast milk was better for children and that few mothers had access to clean water or could afford the expensive commodity. Viola's cancellation of puberty might also prefigure this colonial blockage of bodily flow. See Sasson, 'Milking the Third World: Humanitarianism, Capitalism, and the Moral Economy of the Nestlé Boycott.' *The American Historical Review* 121, 4 (2016): 1196- 1224 (see esp. 1198-99, 1204).
46. According to Ah-King and Hayward, '[m]any species have environmental sex determination, in which temperature, pH, or social environment' 'influence an individual's sex' such that 'sex- changing, intersexuality, and expressing characteristics of both sexes, is, for many organisms, part of their species potential.' See 'Toxic Sexes: Perverting Pollution and Queering Hormone Disruption.' *O-Zone: A Journal of Object-Oriented Studies* 1, 1 (2013): 1–12 (15).
47. I draw on the work of Reena Shadaan and Michelle Murphy in 'Endocrine-Disrupting Chemicals (EDCs) and Industrial and Settler Colonial Structures: Towards a Decolonial Feminist Approach.' *Catalyst: Feminism, Theory, Technoscience* 6, 1 (2020): 1-26. Arguing that EDCs affect metabolism and reproductive health, interrupt signals, and damage organ systems and gene expressions, these authors maintain that EDCs are 'materially a form of colonial environmental violence' 'disrupt[ing[ Lands, sovereignty, bodies, and futures' (13).
48. See Tyson 161, 158.
49. See Siobhan Caroll, An Empire of Air and Water: Uncolonizable Space in the British Imagination, 1750–1850 (Philadelphia: U Pennsylvania Press, 2015).
50. Fiona McNeill explores the labors and symbolic significance of these knitters and spinsters in *Poor Women in Shakespeare* (NY: Cambridge UP, 2007), arguing that poor working women constituted the 'infrastructure of England's national economy' (79). I am greatly indebted to McNeill's extraordinary

work. How to provide for 'free' women is Shakespeare's burden in *Twelfth Night*, and just as we can easily see Viola in need a household and a man to employ her, McNeill comments that Malvolio's clock watching seems better suited for a factory than a household. For an examination of the modern female workplace, see Jordynn Jack, 'Acts of Institution: Embodying Feminist Rhetorical Methodologies in Space and Time.' *Rhetorical Review* 28 (2009): 285-303. Jack details how treatment of women workers typically requires specific protocols for handling time, space, daylight and darkness.

51. Ross 17, 23. For additional background, see K. N. Chaudhuri, *The English East India Company: The Study of an Early Joint-Stock Company 1600–1640* (NY: August Kelley, 1965; NY: Taylor & Francis, 1999), esp 4-8, 14.
52. This 17th century ballad is collected in *The Early British Ballad Archive,* Patricia Ferguson, dir. 'A Warning for married Women. / By the Example of Mrs. Jane Renalds, a West-Country Woman, born neer unto Ply- / mouth; who having plighted her troth to a Sea-man, was afterwards Married to / a Carpenter, and at last carried away by a Spirit: the manner how shall be pre- / sently recited.' University of Glasgow Library Euing Ballads 377. EBBA 31991 https://ebba.english.ucsb.edu/ballad/31991/citation.
53. As Bohannon observes, improving female 'flow' in one area almost always decreases flow in another: the World Bank estimates, for instance, that for every four years of education, a woman's fertility is reduced by one birth.
54. For more details about the conquest of interior space, see Will Jackson, 'The Private Lives of Empire: Emotion, Intimacy, and Colonial Rule.' *Itinerario* 42, 1 (2018): 1-15; and Alfred Archer and Benjamin Matheson, 'Emotional Imperialism.' *Philosophical Topics* 51 (2023): 7-25. Brayton argues that 'somatic space is interarticulated with commercial space' on the seas (144).

    See also Natalie Fixmer-Ortiz, *Homeland Maternity: US Security Culture and the New Reproductive Regime* (Champaign, IL: U Illinois Press, 2019), where she describes how reproductive politics and female health are policed in the 21st century US, its oversight perhaps not so different from the attentions of Orsino's messengers in Shakespeare's Illyria (9-13).

## Disclosure statement

No potential conflict of interest was reported by the author(s).